

__A03__DataExploration.Rmd

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the sub-command to read strings in as factors.

```
#Import packages
library(dplyr)
library(ggplot2)
library(here)
library(lubridate)

#check workspace
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```

here()

## [1] "/home/guest/EDE_Fall2025"

#upload datasets

library(here)

#ECOTOX neonicotinoid dataset

Neonics <- read.csv(
  file = here('Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv'),
  stringsAsFactors = TRUE
)

#Niwot Ridge NEON dataset

Litter <- read.csv(
  file = here('Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv'),
  stringsAsFactors = TRUE
)

```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: Insects are vital to ecological systems, through polination and symbiotic relationships among other things. Neonicotinoids are a poisonous, neurotoxin that are sprayed on crops to deter insects. This chemical can kill insects, but also bees and butterflies. Studying the ecotoxicology of neonicotinoids on insects is relevant because we might be able to weigh the benefits of yielding more crops to the pitfalls of losing pollinators.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: Studying wood and litter from trees can help show us how healthy a forest ecosystem is. Wood and litter are important organic matters, which store carbon and recycle nutrients back into the soil.

4. How is litter and woody debris sampled as part of the NEON network? Read the `NEON_Litterfall_UserGuide.pdf` document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1.Collection method: Litter and fine woody debris are collected from elevated and ground traps 2.Definition of material: lieter is defined as material that is dropped from the forest canopy and has a but end diameter <2cm and a length <50 cm; this material is collected in elevated 0.5m2 PVC traps. Fine wood debris is defined as material that is dropped from the forest canopy and has a but end diameter <2cm and a length >50 cm; 3. Sampling types: Target sampling frequency for elevated traps varies by vegetation present at the site, frequent sampling (1x every 2weeks) in deciduous forest sites during senescence, infrequent year-round sampling (1x every 1-2 months) at evergreen sites

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
#number of colomms
ncol(Neonics)
```

```
## [1] 30
```

```
#number of rows
nrow(Neonics)
```

```
## [1] 4623
```

```
#rows and colomms
dim(Neonics)
```

```
## [1] 4623    30
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
#Factor data because this is a qualitative variable
```

```
Neonics$Effect <- factor (Neonics$Effect)
```

```
#Get a summary of the statistics of end effects on insects due to insecticide
summary(Neonics$Effect)
```

```
##      Accumulation      Avoidance      Behavior      Biochemistry
##           12           102           360           11
##      Cell(s)      Development      Enzyme(s)      Feeding behavior
##           9           136           62           255
##      Genetics      Growth      Histology      Hormone(s)
##          82           38           5           1
##      Immunological      Intoxication      Morphology      Mortality
##          16           12           22           1493
##      Physiology      Population      Reproduction
##           7           1803           197
```

```
#Used sort command to list values in orders of magnitude
effect <- sort(table(Neonics$Effect), decreasing = TRUE)
effect
```

```
##
##      Population      Mortality      Behavior Feeding behavior
##      1803           1493           360           255
##      Reproduction    Development    Avoidance      Genetics
##      197            136            102            82
##      Enzyme(s)       Growth         Morphology     Immunological
##      62             38             22             16
##      Accumulation    Intoxication    Biochemistry    Cell(s)
##      12             12             11             9
##      Physiology      Histology      Hormone(s)
##      7              5              1
```

Answer: This data shows the impact of Neonicotinoids on Insects. The effects of the neurotoxin on the insects is crucial in determining how to move forward in using this chemical. The effect with the greatest frequency also shows us how detrimental the chemical is to them, and how the impact will be reflected and translated into future agriculture crop yield.

- Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed.[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
summary(Neonics$Species.Common.Name, maxsum=6)
```

```
##      Honey Bee      Parasitic Wasp Buff Tailed Bumblebee
##      667           285           183
##      Carniolan Honey Bee      Bumble Bee      (Other)
##      152           140           3196
```

#Honey Bee, Parasitic Wasp and Buff Tailed Bumblebee are the most common insects that are studied. This

Answer: This data shows the impact of neonicotinoids on insects. This metric is relevant because it shows the impact of this chemical on health and survival of insects. If we can understand the most common way insects are impacted by this chemical, we can understand the broader ecological impacts. Results like these are crucial in determining agricultural plot yields and ecological impacts.

- Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class('conco.1..Author')
```

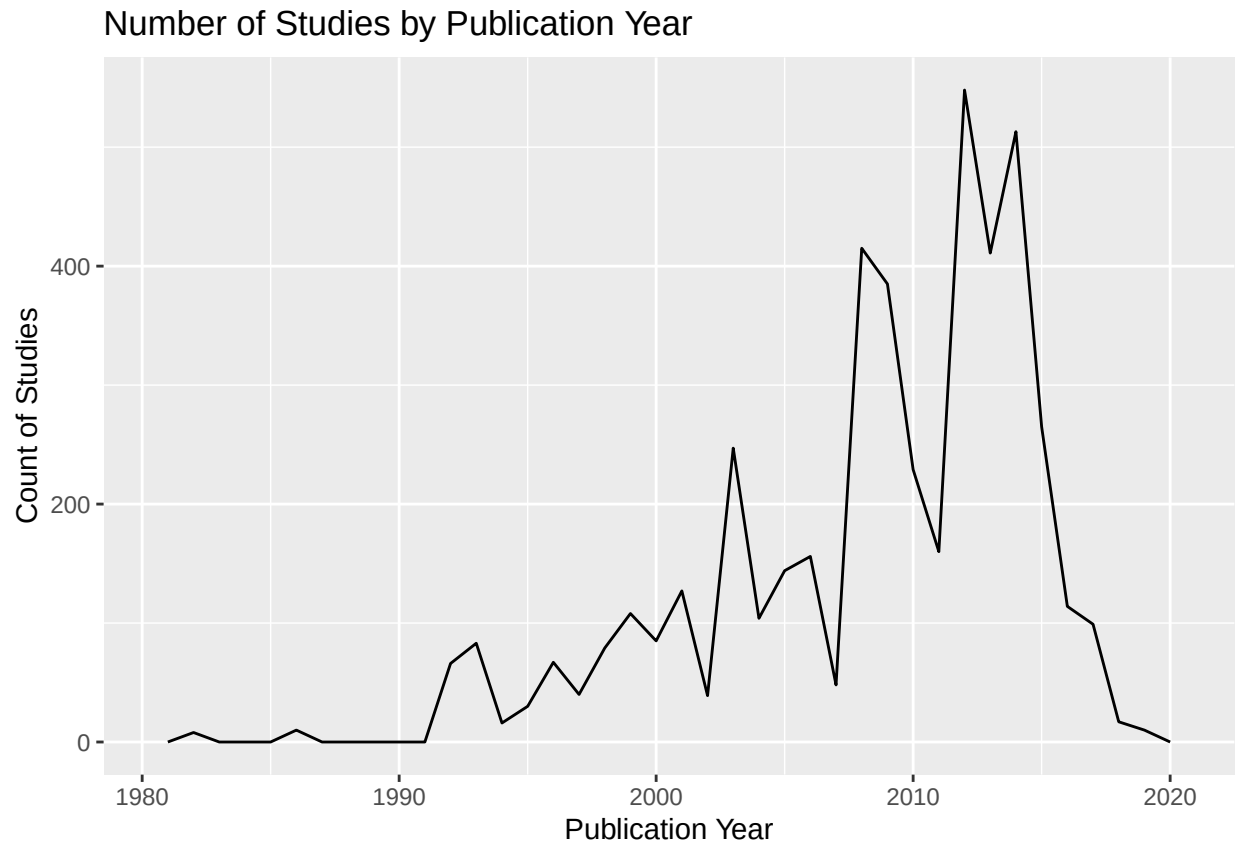
```
## [1] "character"
```

Answer: The `Conc.1.Author` is not numerical, it is stored as a character. This is because it contains quantitative values. These values indicate the treatment conditions- ie whether the substance contained the chemical, the placebo or a mix of the two.

Explore your data graphically (Neonics)

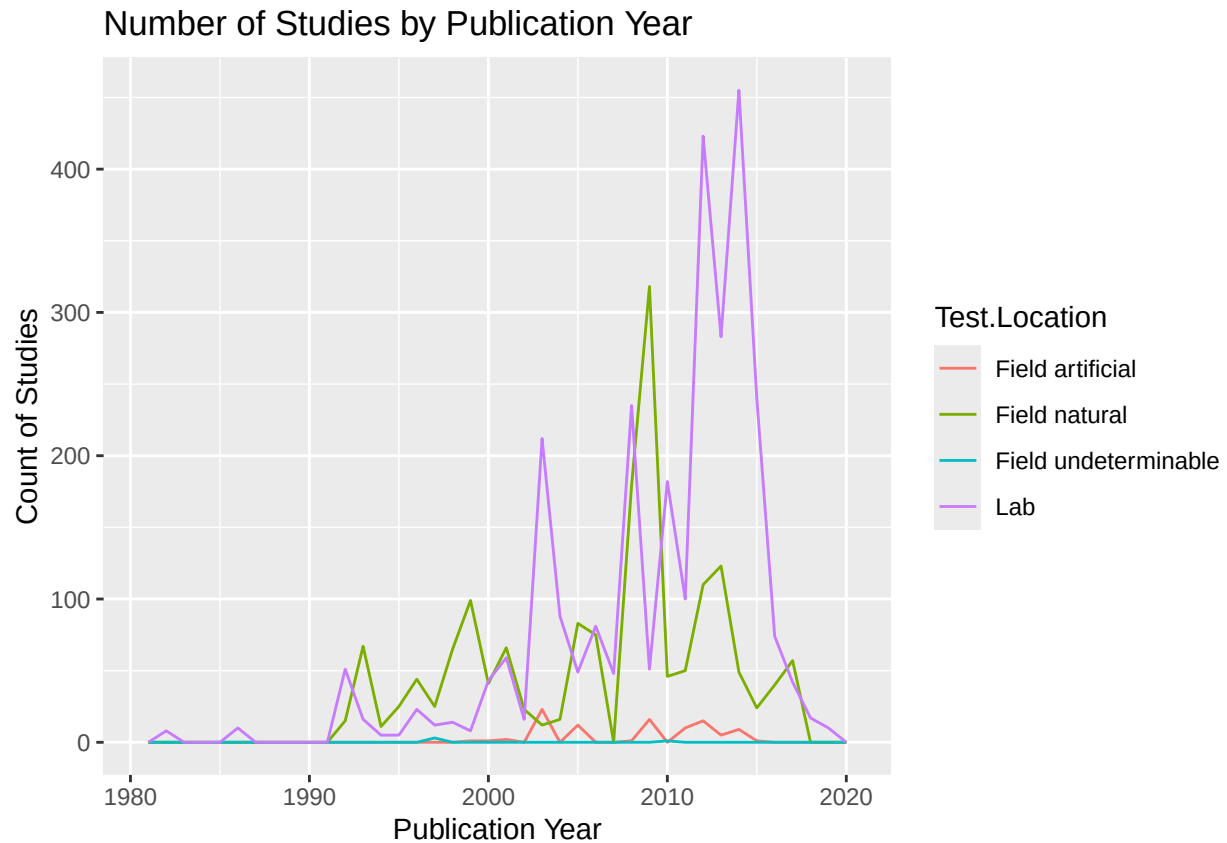
9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
ggplot(Neonics, aes(x = Publication.Year)) +  
  geom_freqpoly(binwidth = 1) +  
  labs(title = "Number of Studies by Publication Year",  
        x = "Publication Year",  
        y = "Count of Studies")
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
#added color line to the bar chart using my notes on how to add color!  
  
ggplot(Neonics, aes(x = Publication.Year, color=Test.Location)) +  
  geom_freqpoly(binwidth = 1) +  
  labs(title = "Number of Studies by Publication Year",  
        x = "Publication Year",  
        y = "Count of Studies")
```



Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: The most common test locations are in the field and the lab. As the years go on, the lab was used more often to test. This trend might be because researchers preferred a more controlled environment to minimize confounding variables.

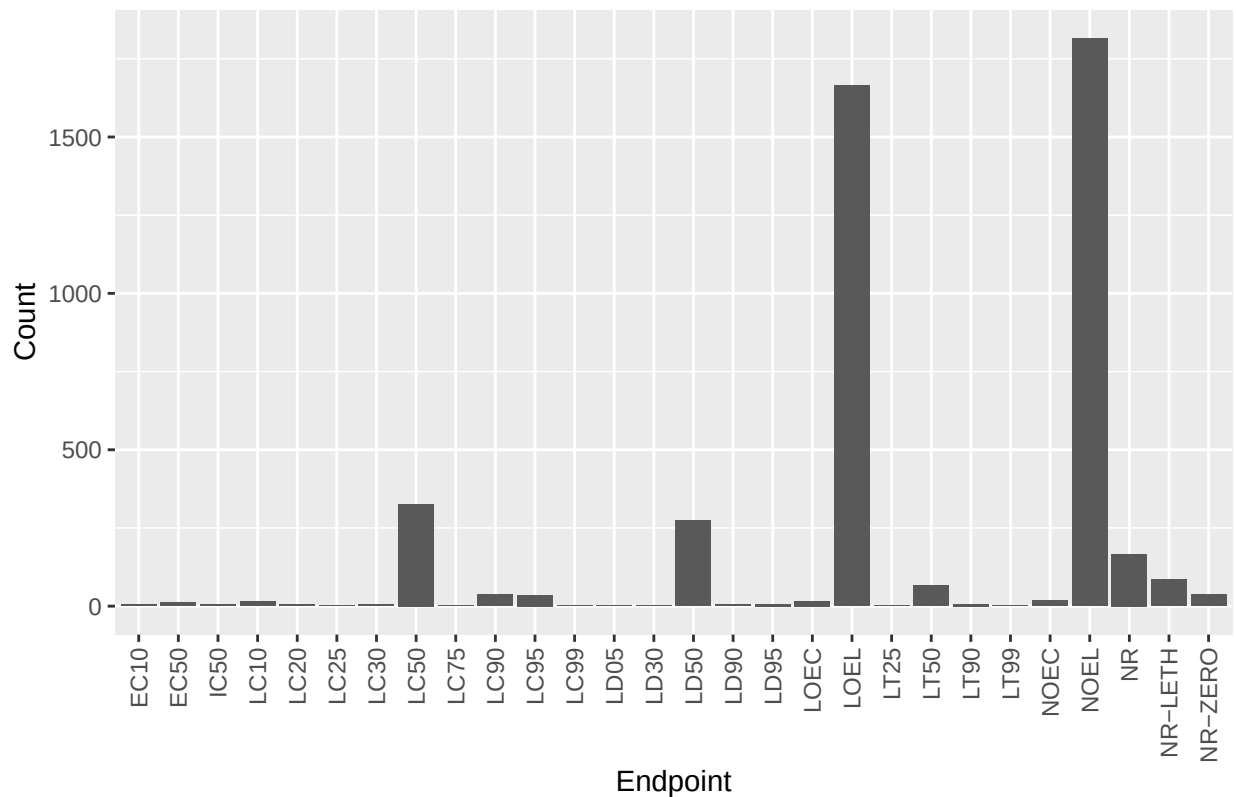
11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
#bar graph of endpoints, assign the x axis to the endpoint value

ggplot(Neonics, aes(x = Endpoint)) +
  geom_bar() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1)) +
  labs(title = "Endpoint in Neonicotinoid Studies",
       x = "Endpoint",
       y = "Count")
```

Endpoint in Neonicotinoid Studies



Answer: NOEL and LOEL are the two most common. According to the PDF, LOEL is “Lowest-observable-effect-level: lowest dose (concentration) producing effects that were significantly different (as reported by authors) from responses of controls (LOEAL/LOEC)”. According to the PDF, NOEL has “No-observable-effect-level: highest dose (concentration) producing effects not significantly different from responses of controls according to author’s reported statistical test (NOEAL/NOEC)”.

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
#Determine class of collect date- it is a character
class(Litter$collectDate)
```

```
## [1] "character"
```

```
#change to a date
Litter$collectDate <- as.Date(Litter$collectDate, format = '%Y-%m-%d')

#confirm new class
class(Litter$collectDate)
```

```
## [1] "Date"
```

```
#use the unique function to determine which dates litter was sampled in August 2018
```

```
unique(Litter$collectDate[format(Litter$collectDate, '%Y-%m') == '2018-08'])
```

```
## [1] "2018-08-02" "2018-08-30"
```

```
# dates when the litter was sampled in August 2018 are 2018-08-02 and 2018-08-30
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
#different plotIDs sampled at Niwot Ridge- 12 Levels
```

```
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
```

```
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
```

```
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

```
# Summary calculation
```

```
summary(Litter$plotID)
```

```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061
```

```
##      20      19      18      15      14      8      16      17
```

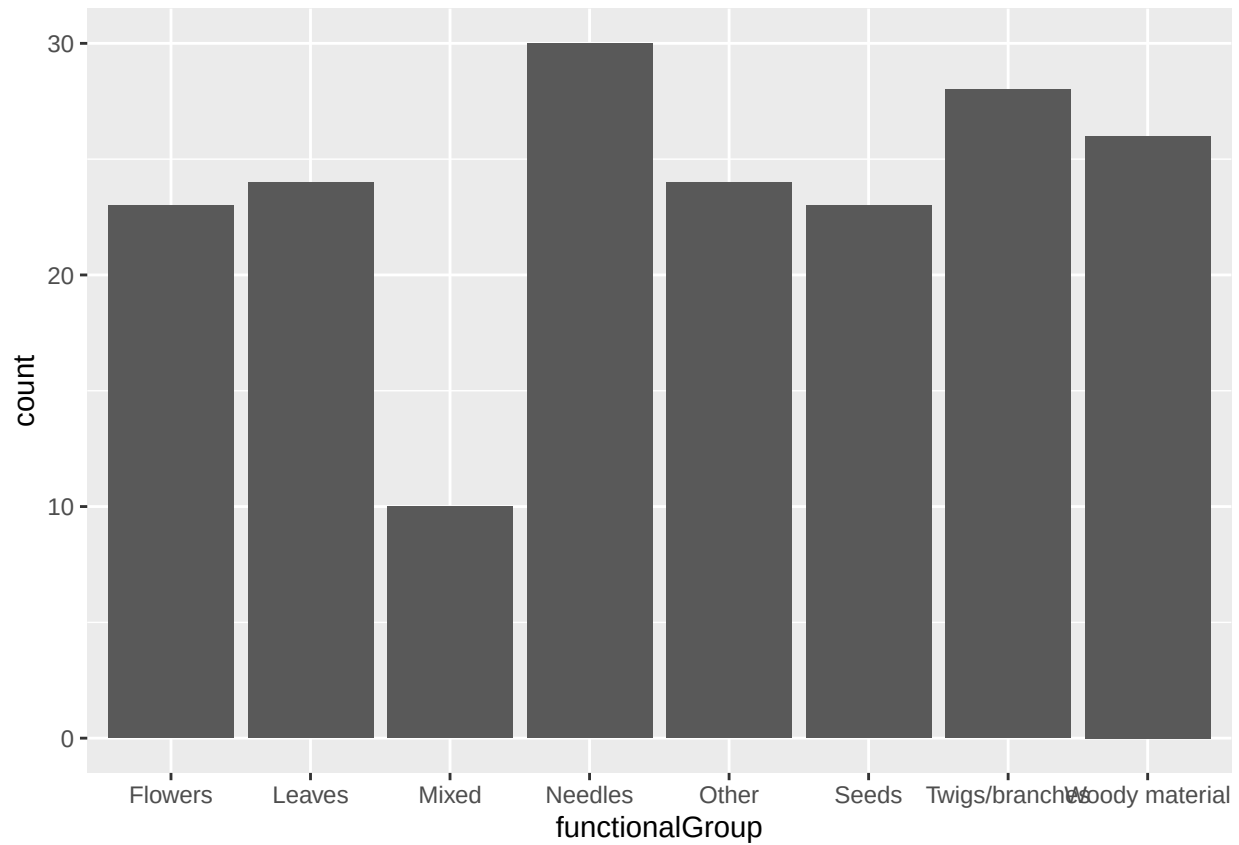
```
## NIWO_062 NIWO_063 NIWO_064 NIWO_067
```

```
##      14      14      16      17
```

Answer: The `unique` plotID shows how many different plotIDs there are, while the `summary` function shows you the frequency, or how many times a plot ID was used.

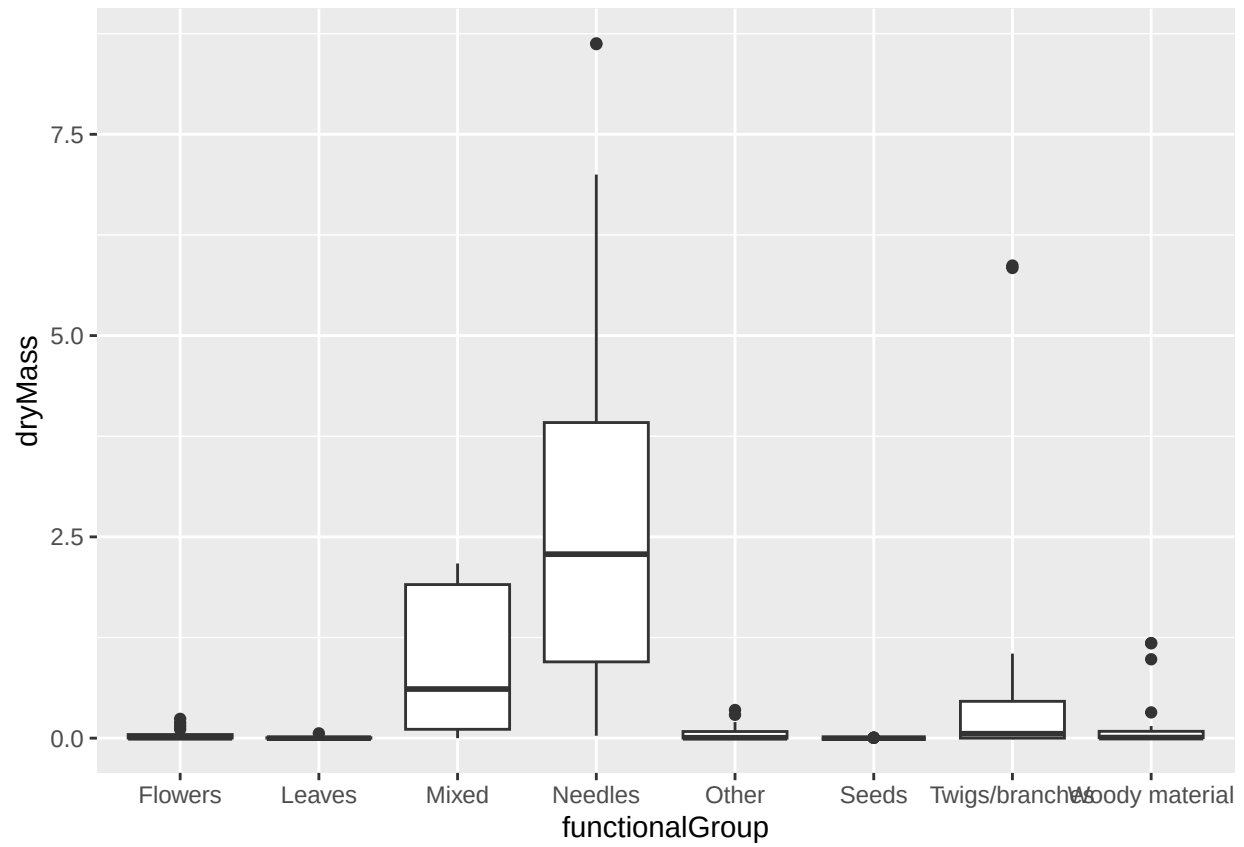
14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
ggplot(Litter, aes(x = functionalGroup)) +  
  geom_bar()
```

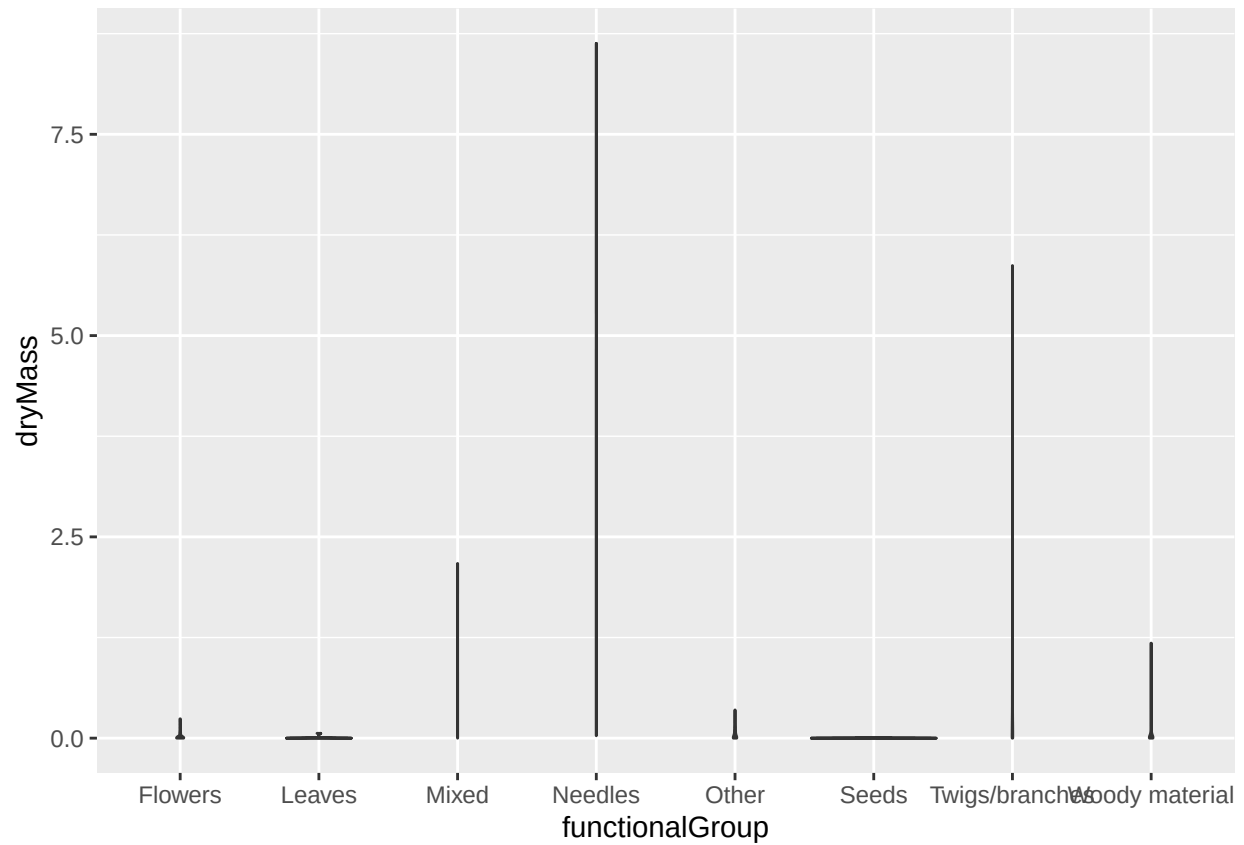



15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
ggplot(Litter) +
  geom_boxplot (
    aes(
      x = functionalGroup,
      y = dryMass,
      group = cut_width(functionalGroup, 1)
    )
  )
```



```
ggplot(Litter) +
  geom_violin(aes(x = functionalGroup, y = dryMass),
    draw_quantiles = c(0.25, 0.5, 0.75))
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: It shows you the key data points like median, mean, upper quartile and lower quartile.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: The needle and mixed litter has the highest biomass.